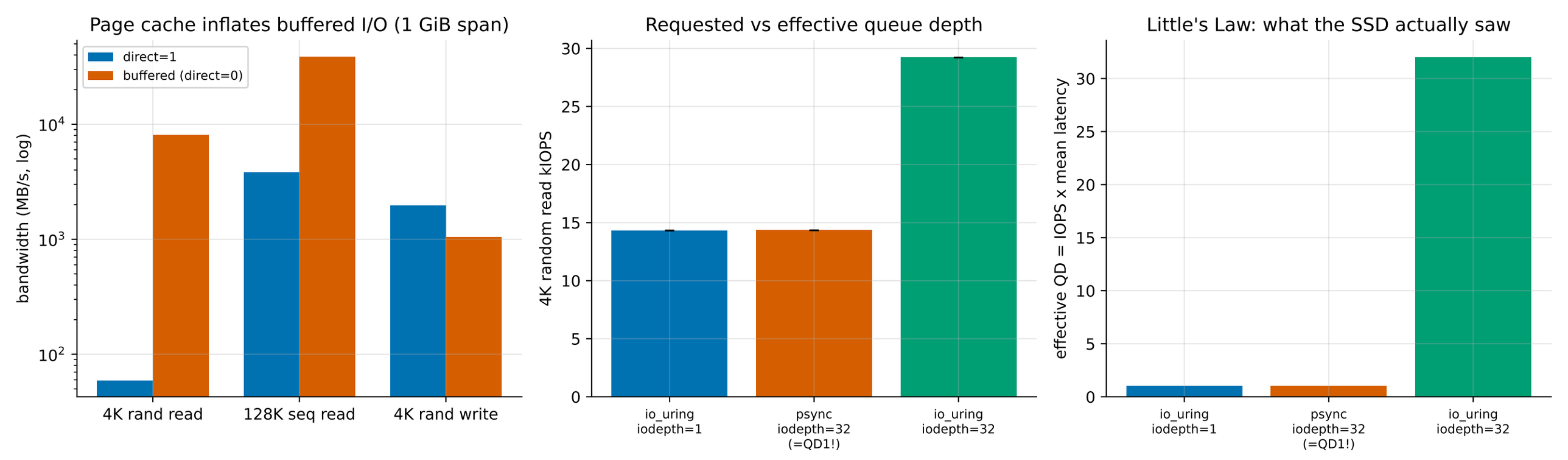
Page cache inflates buffered I/O (1 GiB span)
direct=1
buffered (direct=0)
bandwidth (MB/s, log)
10^4
10^3
10^2
4K rand read
128K seq read
4K rand write
Requested vs effective queue depth
4K random read kIOPS
30
25
20
15
10
5
0
io_uring
iodepth=1
psync
iodepth=32
(=QD1!)
io_uring
iodepth=32
Little's Law: what the SSD actually saw
effective QD = IOPS x mean latency
30
25
20
15
10
5
0
io_uring
iodepth=1
psync
iodepth=32
(=QD1!)
io_uring
iodepth=32